EUV Lithography

From Light to Nanostructure

A Textbook for Engineers and Physicists

Flo (Flowbudget)

July 9, 2026

Extreme Ultraviolet Lithography

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Contents

List of Figures

List of Tables

Preface

Extreme Ultraviolet Lithography (EUV lithography) is the key technology that enables the continuation of Moore's Law into the era of 3 nm and 2 nm nodes. What was once considered physically nearly impossible – generating light at 13,5 nm wavelength, focusing it, and making it usable for mass production – is now industrial reality at TSMC, Samsung, Intel, and ASML.

This textbook covers the physical fundamentals, simulation methods, and process technology of EUV lithography. The structure follows the physical light path: from the plasma source through the collector optics, the mask (multilayer mirror, absorber, pellicle), the projection optics (multilayer-coated mirrors) into the photoresist (chemical amplification, acid diffusion, development). Each chapter connects theory, formulas, numerical methods, and practical examples.

I thank the open-source community (MEEP, S4, PROLITH community) and all those who translate fundamental research into accessible code. Particularly inspiring were the works of *Burnett & Harwood* (EUV Lithography), *Vivek Bakshi* (EUV Sources for Lithography), and the SPIE tutorial series.

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Part I

Fundamentals of EUV Lithography

Chapter 1

Introduction: Why EUV?

1.1 Historical Context: Moore's Law and the Wavelength Spiral

Since Gordon Moore's 1965 prediction, transistor density on chips has doubled every 18–24 months. Optical lithography followed this trend by steadily shortening the exposure wavelength:

Generation	λ	Node (approx.)
G-Line / I-Line (Hg)	436 nm / 365 nm	1 µm – 0,5 µm
KrF Excimer	248 nm	250 nm – 130 nm
ArF Excimer (DUV)	193 nm	130 nm – 7 nm
ArF + Immersion (NA 1.35)	193 nm	45 nm – 7 nm
ArF + Multi-Patterning (SAQP)	193 nm	7 nm – 5 nm
EUV (Extreme Ultraviolet)	13,5 nm	7 nm – 2 nm **and below**

The jump from 193 nm to 13,5 nm (factor 14.3) is the largest wavelength jump in semiconductor manufacturing history. It required not only new light sources but a **completely new optics paradigm**: from lenses (transmission) to mirrors (reflection), from air to vacuum, from quartz glass to multilayer coatings.

1.2 The Physical Necessity: Resolution and Rayleigh Criterion

The minimum resolvable structure width (Half Pitch, HP) follows the Rayleigh criterion:

$$\mathrm{HP} = k_1 \frac{\lambda}{\mathrm{NA}} \tag{1.1}$$

with $k_1 \approx 0.25$–0.30 (process-dependent), λ = wavelength, NA = numerical aperture.

Table 1.1: Theoretical resolution at various λ/NA

System	λ	NA	HP ($k_1 = 0.28$)
DUV Immersion	193 nm	1.35	40 nm
DUV + SAQP	193 nm	1.35	10 nm (effective, 4× patterning)
EUV Low-NA	13,5 nm	0.33	11,4 nm
EUV High-NA	13,5 nm	0.55	6,9 nm

EUV Low-NA (NA 0.33) achieves **single-exposure** resolution that with DUV requires complex multi-patterning (SAQP, 4 exposures + etch steps per layer). This eliminates overlay errors between patterning steps and massively reduces process complexity.

1.3 Challenges of EUV Introduction

1. **No transparent materials** at 13,5 nm → reflective optics (mirrors), multilayer coatings.
2. **Strong absorption in matter** → vacuum in scanner ($\sim$ 1 mbar), no lenses, no conventional masks (pellicle critical).
3. **High-power source needed** → Laser-Produced Plasma (LPP) with 250 W–500 W in 2% BW at wafer.
4. **Stochastic effects** → few photons per feature → photon shot noise, LER, stochastic defects.
5. **Cost** → scanner ~150–300 , High-NA ~350 . High operating costs (source, vacuum, hydrogen, metrology).

1.4 Structure of This Book

Part I Fundamentals: light source, mask, optics, resist physics.

Part II Simulation: RCWA, TMM, aerial image, resist models, stochasticity.

Part III Process: OPC, ILT, process window, metrology, defects, yield.

Chapter 2

The EUV Light Source: Laser-Produced Plasma (LPP)

2.1 Physics of the LPP Source

A 10,6 µm or 1 µm CO_2 / Nd:YAG laser hits tin droplets ($\varnothing \sim 30\,\mu\mathrm{m}$, 50 kHz–80 kHz). The laser ionizes and heats the tin to plasma ($T_e \sim 20\,\mathrm{eV}$–$40\,\mathrm{eV}$). In the plasma, $\mathrm{Sn}^{8+} - -Sn^{14+}ionsemitstronglinesinthe13{,}5\,\mathrm{nm}range(4d - -4f, 4p - -4dtransitions)$.

$$\text{CE (Conversion Efficiency)} = \frac{P_{\text{EUV (2\% BW)}}}{P_{\text{Laser}}} \approx 3\text{–}5\% \tag{2.1}$$

For 250 W EUV at wafer (after optics transmission ~ 2–4%), 20 kW–50 kW laser power is needed (pre-pulse + main-pulse scheme).

2.2 Source Geometry and Collector

- **Pre-pulse** (ns, low energy): deforms droplet to disc/pancake → larger cross-section.
- **Main-pulse** (ps–ns, high energy): generates hot plasma.
- **Collector**: elliptical mirror (Mo/Si multilayer, Ru cap), collects EUV from 2π steradians. 1st focus: plasma. 2nd focus: intermediate focus (IF).
- **Debris mitigation**: $H2$ gas (~ 1 mbar) reacts with Sn vapor to volatile $SnH4$, protects collector. Magnetic field deflects ions.

2.3 Alternative: Discharge-Produced Plasma (DPP)

Z-pinch discharge in Xe/Sn vapor. More compact, no lasers, but:

- Lower CE (~ 1–2%).
- Electrode erosion → contamination, short lifetime.
- Worse spectral purity (more OOB light).

Used for metrology/inspection (e.g., Carl Zeiss EUV scanner for mask inspection), not for HVM.

2.4 Source Parameters for Simulation

Table 2.1: Typical LPP source parameters for HVM

Parameter	Value
Center wavelength	13,5 nm
Bandwidth (FWHM, 2% BW)	0,27 nm
Source size (plasma, eff.)	30 µm–50 µm
Divergence (half-angle)	10°–15°
Power in 2% BW at IF	250 W–500 W
Pulse frequency	50 kHz–80 kHz
Pulse duration	10 ns–20 ns
Polarization	Unpolarized
Debris (Sn, ions)	$\sim$ 1 µg/pulse (before collector)

Part II

The EUV Mask System

Chapter 3

EUV Mask: Structure, Physics, and Simulation

3.1 Structure of the EUV Mask

Unlike DUV masks (transmission), the EUV mask is **reflective**:

1. **Substrate**: Low Thermal Expansion Material (LTEM), $152\,\text{mm} \times 152\,\text{mm} \times 6{,}35\,\text{mm}$, CTE $< 0{,}02/\text{K}$.
2. **Multilayer reflector (ML)**: 40–50 Mo/Si bilayers, $R \approx 70\%$.
3. **Capping layer**: $\sim 4\,\text{nm}$ Ru (ruthenium) – oxidation protection, etch stop.
4. **Absorber**: $\sim 50\,\text{nm}$–$70\,\text{nm}$ TaN / $TaBN$ / $TaBO$ ($n \approx 0.92 + i0.04$).
5. **Hardmask**: $\sim 10\,\text{nm}$ Cr / CrN for etching.
6. **Backside coating**: CrN / TaN for electrostatic clamping.

3.2 Reflection and Phase Shift

The mask operates in **off-axis** illumination mode (dipole, quasar, freeform). The angle of incidence on the mask is typically $\theta \approx 6°$ (chief ray).

- **Bright field (absorber removed)**: light reflects at ML $\rightarrow$ phase ϕ_0.
- **Dark field (absorber present)**: light penetrates absorber, reflects at ML-capping interface $\rightarrow$ phase difference $\Delta\phi$.

Ideally $\Delta\phi = \pi$ (180°) for maximum contrast. Real: $\Delta\phi \approx 170° \ldots 180°$ via absorber thickness/optimization.

3.3 Mask Simulation: Rigorous Coupled-Wave Analysis (RCWA)

RCWA (also Fourier Modal Method) is the standard for periodic/quasi-periodic structures. For the EUV mask (lines/space, contact holes), Maxwell's equations are solved in Fourier space.

Basic Equations (TE Polarization, E_y) Maxwell in frequency domain:

$$\nabla \times \left(\frac{1}{\mu}\nabla \times \mathbf{E}\right) - \omega^2 \varepsilon \mathbf{E} = 0 \tag{3.1}$$

Periodic permittivity $\varepsilon(x, z) = \varepsilon(x + \Lambda, z)$ is expanded in Fourier series:

$$\varepsilon(x, z) = \sum_{m=-\infty}^{\infty} \varepsilon_m(z) e^{imKx}, \quad K = \frac{2\pi}{\Lambda} \tag{3.2}$$

The field is expanded analogously: $E_y(x, z) = \sum_m E_m(z)e^{imKx}$. Substitution yields a linear ODE system for the Fourier coefficients $E_m(z)$:

$$\frac{d^2}{dz^2}\mathbf{E}(z) = \mathbf{A}(z)\mathbf{E}(z) \tag{3.3}$$

where $\mathbf{A}$ contains the convolution of ε coefficients with wavevector components. Solution via eigenvalue decomposition in each homogeneous layer, coupling via boundary conditions (S-matrix formulation for numerical stability).

S-Matrix Algorithm (Redheffer Star Product) For L layers, the scattering matrix $\mathbf{S}^{(l)}$ of each layer is computed and cascaded:

$$\mathbf{S}_{\text{total}} = \mathbf{S}^{(1)} \star \mathbf{S}^{(2)} \star \cdots \star \mathbf{S}^{(L)} \tag{3.4}$$

The reflection coefficients r_m (zeroth order $m = 0$ is the main beam) determine the mask contrast.

Note 3.1 (RCWA Convergence for EUV Masks)**.** For sharp absorber edges (rectangular profile), many Fourier orders are needed: $N_{\text{ord}} \geq 25$–51 (corresponding to $2N + 1$ modes). Li factorization (inverse rule) improves convergence at discontinuities.

3.4 Aerial Image Simulation

The aerial image (image in the projection optics' image plane) follows from coherence theory (Hopkins/Abbe):

$$I(x, y) = \sum_{p,q} J_{pq}\, U_p(x, y) U_q^*(x, y) \tag{3.5}$$

where J_{pq} is the coherence matrix of the illumination (source $\otimes$ pupil function) and U_p is the imaging function of the p-th source point (propagated through mask and optics).

In practice: precompute Transmission Cross Coefficients (TCCs), then:

$$I(x, y) = \sum_{m,n} \text{TCC}_{mn}\, M_m(x, y) M_n^*(x, y) \tag{3.6}$$

with M_m = mask Fourier coefficients.

Chapter 4

Pellicle – The EUV Mask's Shield

4.1 Why a Pellicle?

Particles on the mask ($\geq 30\,\text{nm}$) print. Cleaning is risky (multilayer damage). Pellicle = thin membrane $\sim 50\,\text{mm}$ above mask: particles are defocused $\rightarrow$ non-printing.

4.2 Material Requirements

- Transmission $T > 90\%$ at 13,5 nm (two passes: forth + back $\rightarrow T^2 > 81\%$).
- Mechanically stable: $\Delta P \sim 100\,\text{Pa}$ (pressure difference), no resonances $< 1\,\text{kHz}$ (scanner vibration).
- Thermal: $\sim 100\,\text{W}$ EUV power on mask $\rightarrow$ heating, expansion, overlay errors.
- Chemically inert to $H2$, $O2$, resist outgassing.

4.3 Candidate Materials

Table 4.1: Pellicle materials comparison

Material	Thickness	T (13.5 nm)	E-modulus	Status
Si (single layer)	20 nm	92%	130 GPa	Reference, but brittle
SiN_x	20 nm	88%	250 GPa	Development (ASML, MIT)
Graphene (multilayer)	10 nm	95%	1 TPa	Research (Samsung, IMEC)
CNT network	50 nm	90%	-	Research
$MoSi_2$ / Si multilayer	30 nm	91%	-	Patented (Canon)

4.4 Thermomechanical Simulation (Example)

Equilibrium temperature rise at 100 W on 112 mm × 112 mm membrane:

$$\Delta T = \frac{P_{\text{abs}}}{hA} \approx \frac{10\,\text{W}}{10\,\text{W}/(\text{m}^2\,\text{K}) \times 0{,}025\,\text{m}^2} \approx 40\,\text{K} \tag{4.1}$$

($P_{\text{abs}} \approx 10\%$ of 100 W). Thermal expansion $\Delta L = \alpha L \Delta T$ must be $< 1\,\text{nm}$ (overlay budget) $\rightarrow \alpha < 0{,}1/\text{K}$ (Si: $2{,}6 \cdot 10^{-6}/\text{K}$ – OK, but stress from clamping).

Part III

Projection Optics and Aberrations

Chapter 5

EUV Projection Optics: Mirrors, NA, and Aberrations

5.1 Mirror Optics Instead of Lenses

All 6–10 mirrors are aspheric (freeform, Zernike polynomials), coated with Mo/Si multilayer. No transmissive elements.

Table 5.1: Comparison: Low-NA (0.33) vs. High-NA (0.55) EUV scanners

Parameter	Low-NA (NXE:3400/3600)	High-NA (NXE:5000/520
Numerical aperture (NA)	0.33	0.55
Number of mirrors	6 (4 concave, 2 convex)	8–10 (incl. apodization)
Half-angle (max)	19,3°	33,4°
Resolution (Rayleigh, $k_1 = 0.28$)	11,4 nm HP	6,9 nm HP
Magnification	4×	4× (anamorphic 4×/8×
Image field (ring field)	26 mm × 8,5 mm (curved)	26 mm × 16,5 mm
Anamorphism	No	Yes (4× in X, 8× in Y)
Mirror reflectivity (per mirror)	$\sim 70\%$	$\sim 70\%$ (larger angles)
Total transmission (6 mirrors)	$\sim 11.8\%$	$\sim 5.8\%$ (8 mirrors) / $\sim 4\%$ (10

5.2 Aberration Theory: Zernike Polynomials

Wavefront aberration $W(\rho, \theta)$ over the exit pupil (normalized $\rho \in [0, 1]$):

$$W(\rho, \theta) = \sum_{n,m} Z_n^m(\rho, \theta) \cdot a_n^m \tag{5.1}$$

where Z_n^m are Zernike polynomials (orthonormal on unit circle) and a_n^m are coefficients (in wavelengths or nm). Relevant for EUV:

- Z_4 (defocus): best focus setting, depth of focus $\sim \lambda/\mathrm{NA}^2$.
- Z_5, Z_6 (astigmatism): mirror figure errors, mounting.
- Z_7, Z_8 (coma): field-dependent, asymmetry.
- Z_9 (spherical aberration): higher orders, mirror figure.
- Z_{11}, Z_{12} (trefoil): 3-fold symmetry, polishing errors.

- $Z_{16} - -Z_{22}$ (higher orders): critical for anamorphic High-NA.

Note 5.1 (Aberration Budget High-NA)**.** Total RMS wavefront error (WFE) budget: 0,5 nm–1,0 nm RMS ($\lambda/13$–$\lambda/27$ at 13,5 nm). Individual mirrors: 0,1 nm–0,2 nm RMS.

5.3 Anamorphic Optics (High-NA)

Different magnification in X (4×) and Y (8×) → elliptical pupil, different NA in X/Y ($NA_x \approx 0.55$, $NA_y \approx 0.275$ effective). Consequences:

- Mask patterns must be drawn 2× larger in Y (pre-bias).
- Asymmetric illumination (dipole in Y direction).
- Depth of focus asymmetric: $DoF_y \approx 2 \times DoF_x$.
- Overlay sensitivity doubled in Y.

5.4 Multilayer Mirrors in the Optics

Each mirror has 40–50 Mo/Si pairs, optimized for angle of incidence θ and bandwidth. Reflectivity $R(\theta, \lambda)$ varies across pupil (different θ). Apodization (intentional transmission variation) corrects intensity distribution in pupil.

Part IV

Resist Modeling and Process Simulation

Chapter 6

Chemically Amplified Resist (CAR) – Physics and Chemistry

6.1 Reaction-Diffusion Equations

Exposure → acid generation → post-exposure bake (PEB) → catalytic deprotection → development.

$$\frac{\partial C_{\text{PAG}}}{\partial t} = -\sigma_{\text{PAG}} I(x,y,z) C_{\text{PAG}} \qquad \text{(PAG consumption)} \tag{6.1}$$

$$\frac{\partial C_{\text{Acid}}}{\partial t} = \Phi_{\text{acid}} \sigma_{\text{PAG}} I C_{\text{PAG}} + D_{\text{Acid}} \nabla^2 C_{\text{Acid}} - k_{\text{quench}} C_{\text{Acid}} C_{\text{Base}} \qquad \text{(acid)} \tag{6.2}$$

$$\frac{\partial C_{\text{Prot}}}{\partial t} = -k_{\text{deprot}}(T) C_{\text{Acid}} C_{\text{Prot}} \qquad \text{(deprotection)} \tag{6.3}$$

$$\frac{\partial C_{\text{Base}}}{\partial t} = -k_{\text{quench}} C_{\text{Acid}} C_{\text{Base}} + D_{\text{Base}} \nabla^2 C_{\text{Base}} \qquad \text{(base/quencher)} \tag{6.4}$$

Where:

- $I(x,y,z)$: intensity in resist (from aerial image + Fresnel transmission).
- σ_{PAG}: PAG absorption cross-section ($\sim 1 \cdot 10^{-18}\,\text{cm}^2$).
- Φ_{acid}: acid quantum yield/photon (~ 0.1–0.5).
- D_{Acid}: acid diffusion coefficient ($\sim 10^{-14}\,\text{cm}^2/\text{s}$ at PEB temp).
- $k_{\text{deprot}} = A \exp(-E_a/k_B T)$: Arrhenius equation.

6.2 EUV-Specific Resist Phenomena

6.2.1 Photon Shot Noise

EUV photon energy: $E_{\text{ph}} = hc/\lambda \approx 92\,\text{eV}$. Dose $D = 30\,\text{mJ/cm}^2 \rightarrow N_{\text{ph}} = D/E_{\text{ph}} \approx 2 \cdot 10^{14}/\text{cm}^2$. In a 13 nm HP pixel ($\sim 170\,\text{nm}^2$): ~ 3400 photons. Poisson noise: $\sigma_N/N = 1/\sqrt{N} \approx 1.7\%$. **Dominant contributor to LER/LWR**.

6.2.2 Secondary Electrons

EUV photon → photoelectron ($\sim 80\,\mathrm{eV}$) → cascade → $\sim$ 3–5 secondary electrons per photon (energy $1\,\mathrm{eV}$–$20\,\mathrm{eV}$). These generate acid **non-locally** (broadening $\sim 5\,\mathrm{nm}$–$10\,\mathrm{nm}$). Monte Carlo model (e.g., `SEDREC`, `IMPEL`) needed.

6.2.3 Outgassing / Contamination

Resist outgassing deposits on multilayer mirrors → reflectivity loss. $H2$ ambient in scanner mitigates this.

6.3 Development Model: Mack Model / String Development

Dissolution rate R as function of deprotection $m = C_{\mathrm{Prot}}/C^0_{\mathrm{Prot}}$:

$$R(m) = R_{\max}\frac{m^n}{m^n + m_{50}^n} + R_{\min} \tag{6.5}$$

Typical: $n \approx 1.5$–3, $m_{50} \approx 0.2$–0.4. Development simulated via eikonal equation / level-set / cellular automaton.

Chapter 7

Process Window, OPC, and Inverse Lithography

7.1 Process Window

Combination of focus (z) and dose (D) delivering CDs within specs ($\pm 10\,\%$) and overlay ($< 1{,}5\,\mathrm{nm}$).

- **DoF (Depth of Focus)**: focus range at nominal dose.
- **EL (Exposure Latitude)**: dose range at best focus.
- **MEEF (Mask Error Enhancement Factor)**: $\Delta CD_{\mathrm{wafer}}/\Delta CD_{\mathrm{mask}}$. Typical 3–5 for EUV.

7.2 OPC (Optical Proximity Correction)

Model-based OPC: iterative correction of mask (M) so wafer CD (W) hits target.

$$M^{(k+1)} = M^{(k)} + \alpha \left(W_{\mathrm{target}} - W(M^{(k)})\right) \cdot \mathrm{MEEF}^{-1} \tag{7.1}$$

- **Rule-based OPC**: serifs, hammerheads, bias rules. Fast, for dense patterns.
- **Model-based OPC**: rigorous simulation (aerial image + resist) per iteration. Slow, accurate.
- **ILT (Inverse Lithography Technology)**: mask as continuous variable (grayscale), optimization via gradient descent / AD (automatic differentiation). Produces curved, unorthodox mask shapes – better process window.

7.3 Mask 3D Effects (M3D)

Absorber height ($\sim 60\,\mathrm{nm}$) $\neq 0$ $\rightarrow$ phase differences between diffraction orders, best focus shift (Best Focus Shift – BFS) dependent on pattern orientation (H vs. V). Correction: **M3D-OPC**, **mask bias**, **undercut** during etch.

7.4 EUV-Specific OPC Challenges

1. **Stochasticity**: shot noise $\rightarrow$ local CD variation $\rightarrow$ OPC cannot apply deterministic correction to stochastic effects.

2. **Flare**: stray light in optics ($\sim$ 1–5%) $\rightarrow$ background exposure, CD bias for isolated lines.
3. **Polarization**: at High-NA strong polarization effects (TE/TM splitting) $\rightarrow$ vector OPC needed.
4. **Anamorphism**: different OPC in X/Y.

Part V

Metrology, Defects, and Yield

Chapter 8

EUV Metrology: CD, Overlay, Defect Inspection

8.1 CD Measurement (Critical Dimension)

- **CD-SEM**: standard, but e-beam damage (resist shrinkage, charging). Low-voltage (100 eV–500 eV) + charge compensation.
- **Scatterometry (OCD)**: optical diffraction on periodic structures → regression model → 3D profile (CD, sidewall angle, height). Non-destructive, fast.
- **AFM**: reference measurement, slow.
- **EUV scatterometry**: actinic pattern inspection (see below).

8.2 Overlay Metrology

- **IMB (Image Based Overlay)**: image-based, box-in-box, $\sim 0{,}5$ nm precision.
- **DBO (Diffraction Based Overlay)**: diffraction order, higher precision ($\sim 0{,}2$ nm), but asymmetry-sensitive.
- **EUV overlay**: actinic measurement (same λ as exposure) → eliminates optics aberration difference.

8.3 Defect Inspection: Actinic vs. Non-Actinic

Table 8.1: Mask inspection technologies

Method	Wavelength	Resolution	Status
DUV inspection (193 nm)	193 nm	50 nm	HVM (current)
EUV actinic inspection (ACTIS)	13,5 nm	13 nm HP	Development (ASML, Carl Zeiss
E-beam inspection	e-beam	10 nm	Slow, review
Ptychography (lab)	EUV / X-ray	<10 nm	Research

Actinic inspection (ACTIS): uses EUV light ($\lambda = 13{,}5$ nm) → sees only defects that print (printability). No false alarms from non-printing defects. Throughput target: 10 mm/s^2.

8.4 Pellicle Defects

Particles on pellicle → defocus → contrast loss. Printability threshold:

$$d_{\text{crit}} \approx 0.5 \times \text{HP} \times \sqrt{\frac{\Delta z}{\text{DoF}}} \tag{8.1}$$

with Δz = pellicle-mask distance ($\sim 50\,\text{mm}$), DoF = depth of focus ($\sim 50\,\text{nm}$ at NA 0.33). For HP 13 nm: $d_{\text{crit}} \approx 150\,\text{nm}$ on pellicle (vs. 30 nm on mask).

Chapter 9

Stochastic Effects, LER, and Yield

9.1 Sources of Stochasticity

1. **Photon shot noise**: Poisson statistics of EUV photons.
2. **PAG statistics**: discrete PAG molecules ($\sim 1 \cdot 10^{20}/\text{cm}^3 \to \sim 1700$ molecules in 13 nm HP voxel).
3. **Acid diffusion**: random walk $\to$ blur.
4. **Development noise**: molecular nature of developer.
5. **Mask roughness (LWR)**: transfers to wafer (MEEF $\times$).

9.2 Line Edge Roughness (LER) / Line Width Roughness (LWR)

LER = edge roughness (PSD: power spectral density). LWR = width variation (correlated with LER).

$$\text{LWR}^2 \approx 2 \times \text{LER}^2 \quad \text{(for uncorrelated edges)} \tag{9.1}$$

Target: LWR $< 1{,}5$ nm (3σ) at HP 13 nm.

9.3 Modeling: Monte Carlo vs. Analytical

- **Monte Carlo (e.g., `PROLITH` Stochastic, `SEMulator3D`)**: tracks individual photons, electrons, molecules. Slow, but physically complete.
- **Analytical (variance propagation)**: $\sigma^2_{\text{CD}} = \sum(\partial CD/\partial x_i)^2 \sigma^2_{x_i}$. Fast, for OPC loops.
- **Hybrid**: MC for calibration, analytical for production.

9.4 Stochastic Defects: Microbridging, Broken Lines, Missing Contacts

Defect probability per feature:

$$P_{\text{defect}} \approx \frac{1}{2}\text{erfc}\left(\frac{\Delta_{\text{margin}}}{\sqrt{2}\sigma_{\text{total}}}\right) \tag{9.2}$$

Δ_{margin}: process window margin to defect threshold. σ_{total}: total stochasticity.

At $30\,\text{mJ/cm}^2$, HP $13\,\text{nm}$: $\sigma_{\text{CD}} \approx 1{,}0\,\text{nm} \to P_{\text{bridge}} \sim 10^{-3}$–$10^{-4}$ per μm line. For chip with $1 \cdot 10^9$ contacts: yield killer.

9.5 Mitigation

- Higher dose ($\uparrow D \to \downarrow \sigma_{\text{shot}}$) $\to$ but throughput $\downarrow$, outgassing $\uparrow$.
- Better resists (higher Φ_{acid}, lower D_{Acid}, higher n in Mack model).
- EUV-specific OPC (stochastic-aware OPC: local dose modulation, assist features).
- Self-assembly (DSA) as amplifier.
- High-NA: smaller features $\to$ fewer photons per feature $\to$ **worse**! (dose must increase).

Appendix A

Formula Collection: Key Equations of EUV Lithography

A.1 Imaging

$$\text{Half-Pitch (Rayleigh)} \quad HP = k_1 \frac{\lambda}{\mathrm{NA}} \tag{A.1}$$

$$\text{Depth of Focus} \quad DoF = k_2 \frac{\lambda}{\mathrm{NA}^2} \tag{A.2}$$

$$\text{MEEF} \quad \mathrm{MEEF} = \frac{\Delta CD_{\text{wafer}}}{\Delta CD_{\text{mask}}} \approx \frac{1}{\mathrm{NILS}} \frac{HP}{CD} \tag{A.3}$$

$$\text{NILS} \quad \mathrm{NILS} = \frac{1}{I_{\max}} \left| \frac{dI}{dx} \right|_{x=CD/2} \tag{A.4}$$

$$\text{Image Log Slope (ILS)} \quad \mathrm{ILS} = \left| \frac{d \log I}{dx} \right|_{x=CD/2} \tag{A.5}$$

A.2 Multilayer Reflector

$$\text{Bragg Condition} \quad 2d \sin\theta = m\lambda \tag{A.6}$$

$$\text{Optimal d-spacing (Mo/Si, } m = 1,\ \theta \approx 6^\circ) \quad d \approx \frac{\lambda}{2\sin\theta} \approx 6{,}9\,\mathrm{nm} \tag{A.7}$$

$$\text{Reflectivity (kinematic, } N \text{ pairs)} \quad R \approx \left(\frac{\Delta n}{2\bar{n}} \right)^2 N^2 \quad (\text{for small } \Delta n) \tag{A.8}$$

$$\text{Bandwidth (FWHM)} \quad \frac{\Delta\lambda}{\lambda} \approx \frac{2}{\pi N} \frac{\Delta n}{\bar{n}} \tag{A.9}$$

A.3 Resist (CAR)

$$\text{Acid generation} \quad \frac{d[\mathrm{Acid}]}{dt} = \Phi \sigma I [\mathrm{PAG}] \tag{A.10}$$

$$\text{Diffusion length (PEB)} \quad L_{\text{diff}} = \sqrt{2 D_{\text{Acid}} t_{\text{PEB}}} \tag{A.11}$$

$$\text{Mack development rate} \quad R(m) = R_{\max} \frac{m^n}{m^n + m_{50}^n} + R_{\min} \tag{A.12}$$

$$\text{Shot noise (relative CD variation)} \quad \frac{\sigma_{\mathrm{CD}}}{CD} \approx \frac{1}{\mathrm{NILS} \cdot CD} \sqrt{\frac{1}{N_{\text{ph}}}} \tag{A.13}$$

A.4 Stochasticity

Photons per pixel $$N_{\mathrm{ph}} = \frac{D \cdot A_{\mathrm{pixel}}}{E_{\mathrm{ph}}} \tag{A.14}$$

Poisson noise $$\sigma_N = \sqrt{N} \tag{A.15}$$

LER (simple model) $$\mathrm{LER}_{3\sigma} \approx \frac{3}{\mathrm{NILS}} \sqrt{\frac{1}{N_{\mathrm{ph}}} + \frac{1}{N_{\mathrm{PAG}}} + \left(\frac{L_{\mathrm{diff}}}{CD}\right)^2} \tag{A.16}$$

A.5 Defects

Printability (particle on mask) $$\text{Prints if } d_{\mathrm{particle}} > 0.5 \cdot HP \cdot \sqrt{\frac{z}{\mathrm{DoF}}} \tag{A.17}$$

Printability (particle on pellicle) $$\text{Prints if } d_{\mathrm{pell}} > 0.5 \cdot HP \cdot \sqrt{\frac{z_{\mathrm{pell}}}{\mathrm{DoF}}} \tag{A.18}$$

Flare (veiling glare) $$I_{\mathrm{total}} = I_{\mathrm{aerial}} + \eta_{\mathrm{flare}} \cdot \langle I_{\mathrm{aerial}} \rangle \tag{A.19}$$

Appendix B

Material Database: Optical Constants at 13,5 nm

Table B.1: Refractive indices $n = 1 - \delta + i\beta$ at 13,5 nm (CXRO / Henke data)

Material	δ (10^{-3})	β (10^{-3})	ρ (g/cm^3)
Mo	4.61	2.12	10.2
Si	3.85	0.31	2.33
Ru	4.12	2.85	12.4
TaN	3.95	1.85	13.5
TaBN	3.88	1.72	12.8
TaBO	3.75	1.55	11.2
SiO_2	3.12	0.18	2.2
SiN	3.45	0.25	2.9
C (graphite)	2.85	0.42	2.2
Hydrogen (gas)	~ 0	~ 0	0.00009

Appendix C

Abbreviations and Glossary

ACTIS Actinic Pattern Inspection (aktinische Maskeninspektion)

BEUV Beyond EUV

BFS Best Focus Shift

CAR Chemically Amplified Resist

CD Critical Dimension

CDU CD Uniformity

DBO Diffraction Based Overlay

DoF Depth of Focus

DPP Discharge Produced Plasma

DSA Directed Self Assembly

EL Exposure Latitude

EUV Extreme Ultraviolet

HHG High Harmonic Generation

HP Half Pitch

HVM High Volume Manufacturing

ILT Inverse Lithography Technology

IMB Image Based Overlay

LER Line Edge Roughness

LWR Line Width Roughness

LPP Laser Produced Plasma

M3D Mask 3D Effects

MEEF Mask Error Enhancement Factor

ML Multilayer

NA Numerical Aperture

NILS Normalized Image Log Slope

OCD Optical Critical Dimension (Scatterometrie)

OPC Optical Proximity Correction

PAG Photo Acid Generator

PEB Post Exposure Bake

PSD Power Spectral Density

PVD Physical Vapor Deposition

RCWA Rigorous Coupled Wave Analysis

SEM Scanning Electron Microscope

SRAF Sub-Resolution Assist Feature

TCC Transmission Cross Coefficient

TMM Transfer Matrix Method

WFE Wavefront Error

Zernike Zernike Polynomials (Aberrationsbasis)

Appendix D

References and Further Reading

D.1 Standard Works

1. **Burnett, J. H. & Vandervorst, W. (eds.)**: *EUV Lithography*, SPIE Press, 2nd edition, 2019. (*The* standard work, 20+ authors).

2. **Vivek Bakshi (ed.)**: *EUV Lithography*, SPIE Tutorial Texts TT98, 2018.

3. **Mack, C. A.**: *Fundamental Principles of Optical Lithography*, Wiley, 2007. (Basis for resist/aerial image).

4. **Rutherford, S. et al.**: *Introduction to Microlithography*, SPIE, 2019.

D.2 Key Papers (Selection)

1. **Naulleau et al.**: *Actinic inspection for EUV lithography*, Proc. SPIE 10143, 2017.

2. **Levallois et al.**: *High-NA EUV: Anamorphic optics and mask implications*, Proc. SPIE 10957, 2019.

3. **Erickson et al.**: *Stochastic modeling of EUV resist exposure*, J. Micro/Nanolith. MEMS MOEMS, 2020.

4. **Gallatin et al.**: *EUV resist stochastic effects: Photon shot noise vs. chemical noise*, Proc. SPIE 11324, 2020.

5. **Graindorge et al.**: *Pellicle development for EUV lithography*, J. Vac. Sci. Technol. B, 2021.

6. **Hendrix et al.**: *Machine learning for OPC and ILT*, Proc. SPIE 11327, 2020.

7. **Spence et al.**: *Mask 3D effects in EUV lithography*, Proc. SPIE 9777, 2016.

8. **Burkhardt et al.**: *Source optimization for EUV lithography*, Proc. SPIE 9048, 2014.

D.3 Online Resources

- **CXRO Optical Constants**: `https://henke.lbl.gov/optical_constants/` (reference for n, k).

- **ASML EUV Lithography**: `https://www.asml.com/en/technology/euv-lithography`

- **SPIE Digital Library**: https://www.spiedigitallibrary.org/ (conferences: SPIE Advanced Lithography, EUV Lithography).
- **SEMATECH / SRC Lithography Reports**: https://www.src.org/
- **IMEC EUV Litho**: https://www.imec-int.com/en/euv-lithography